

VASANTHKUMAR N

Mechanical Engineering | B.E. (2023–2027) | Kamaraj College of Engineering & Technology

Email: vasanth19102005@gmail.com • LinkedIn: [linkedin.com/in/vasanthkumarn1910](https://www.linkedin.com/in/vasanthkumarn1910) • Phone: +91 6369797334

Portfolio: <https://vasanth-nm.github.io/portfolio> • Location: Madurai, Tamil Nadu - 625022

OBJECTIVE

Final year B.E. Mechanical Engineering student (Class of 2027) with a strong foundation in CAD modelling, FEA simulation, and IoT-based systems. Seeking an entry-level Design Engineering or Manufacturing Engineer role to apply skills in Fusion, ANSYS, and hands-on fabrication for innovative product development.

EDUCATION

Institution	Course	Percentage / CGPA	Year
Kamaraj College of Engineering & Technology	B.E. Mechanical Engineering	CGPA: 7.75 (Up to semester-5)	2023–2027
Amudham Matric Hr. Sec. School	HSC	71%	2023

TECHNICAL SKILLS

CAD Software	AutoCAD (2D Drafting), PTC Creo (3D Parametric CAD), Autodesk Fusion 360, CATIA V5
Simulation / FEA	ANSYS Workbench (Static Structural, Steady-State Thermal, Modal Analysis)
IIoT & Automation	ESP8266 Wi-Fi, RFID Telemetry, Sensors (ADXL345 Vibration, DS18B20 Temp), Firebase, Python ML
Manufacturing & QC	CNC Programming (G-Code/M-Code), GD&T Tolerancing, Metrology & Inspection
3D & Media Tools	Blender (3D Modeling & Rendering), DaVinci Resolve (Video Production)

PROJECTS

Digital Twin-Based Predictive Maintenance System — Capstone Project (2026–27)

- Developing a Digital Twin framework integrating ANSYS FEA, ESP8266 IoT telemetry, Firebase cloud, Unity 3D, and Machine Learning for real-time health monitoring of rotating machinery.
- Engineered a real-time data acquisition system with ADXL345 3-axis vibration and DS18B20 temperature sensors mounted on motor bearing housings.
- Implementing a Random Forest ML model to detect and classify bearing faults, dynamic imbalance, and shaft misalignment for early failure prevention.
- Designing a synchronized 3D Unity digital twin dashboard displaying live sensor data to deliver an affordable predictive maintenance solution for MSMEs.

Design & Fabrication of a Compact Heat Recovery System — 3rd Year Project (2024–25)

- Designed a shell-and-helical-coil heat exchanger for industrial waste heat recovery in Autodesk Fusion under compact envelope constraints.
- Conducted thermal and static structural stress simulation in ANSYS Workbench to validate fluid velocity, heat transfer coefficients, and shell pressure drops.
- Fabricated and tested a physical working prototype; measured heat transfer effectiveness and fluid flow parameters across varying operating conditions.
- Optimized helical coil pitch geometry and shell dimensions to achieve high thermal effectiveness under compact form-factor constraints.

Smart Non-Newtonian Fluid Dynamic Speed Bump — Thinkathon Innovation, KCET

- Engineered a concept for an adaptive speed bump utilizing shear-thickening non-Newtonian fluid (STF) that yields under low-speed vehicle roll while rigidifying into a solid obstacle upon high-speed impact.
- Conceived a passive mechanism enabling emergency vehicles (ambulances, fire engines) to traverse bumps smoothly without jarring chassis deceleration.
- Designed a reinforced elastomeric chamber housing layout engineered to resist cyclic heavy axle loads and prevent fluid leakage.

INTERNSHIPS & INDUSTRIAL EXPERIENCE

Pentagon Apparels Unit II, Virudhunagar — IoT / RFID Sewing Performance Tracking System

- Implemented an RFID-based shop-floor tracking system using ESP8266 Wi-Fi modules to monitor operator-level sewing output in real time across manufacturing lines.
- Integrated Wi-Fi telemetry with a live centralized web dashboard, enabling plant supervisors to identify production bottlenecks and balance line throughput instantly.
- Eliminated manual paper logging errors and reduced data reconciliation latency across active production shifts.

Swing Setter — Paint Shop Department Internship (15 Days)

- Observed end-to-end industrial finishing operations including chemical surface pre-treatment, primer/topcoat spray application, and oven curing stages.
- Studied quality control checkpoints for finish consistency, coating thickness (DFT), and surface defect diagnosis.
- Gained practical exposure to industrial safety protocols, personal protective equipment (PPE), and workflow sequencing in an active manufacturing plant.

ACHIEVEMENTS & HONORS

- **2nd Prize** — Technical Quiz, Yuvasakthi 2K26 National Level Technical Symposium, NPR College of Engineering & Technology, Natham (Mar 2026).
- **3rd Prize** — SkyForge Event, Euphoria Techno Management Meet, Kalasalingam Academy of Research & Education (Mar 2026).
- **NPTEL Elite Certificate** — Product Design & Manufacturing, IIT Kanpur / SWAYAM (Jan–Apr 2026).

CERTIFICATIONS & PROFESSIONAL TRAINING

- **NPTEL Elite Certificate — Product Design & Manufacturing** | IIT Kanpur / SWAYAM (Jan–Apr 2026)
- **Industrial Internet of Things (IIoT) Hands-on Training** | SASTRA University, Thanjavur
- **CATIA V5 Certification** | Kamaraj College of Engineering & Technology
- **Advanced Creo (3D CAD) Professional Certificate** | Udemy
- **CNC Programming Training (G-Code/M-Code)** | NTTF (Nettur Technical Training Foundation), Bangalore
- **ISTE Student Member (SM No. 11709)** | Indian Society for Technical Education (2025–2027)

RELEVANT COURSEWORK

Design of Machine Elements • Finite Element Analysis (FEA) • Thermodynamics & Heat Transfer • Fluid Mechanics & Machinery • Computer Aided Design & Manufacturing (CAD/CAM) • Industrial Internet of Things & Sensors • Mechatronics & Control Systems • Metrology & Quality Inspection

SOFT SKILLS & DOMAIN INTERESTS

Problem-Solving • Creative Thinking • Cross-Functional Team Collaboration • 3D Parametric Part Modelling • CAD/CAM Automation • Industrial IoT Telemetry

DECLARATION

I hereby solemnly declare that all the information furnished above is true, complete, and correct to the best of my knowledge and belief.

Date: _____

Place: Madurai, Tamil Nadu

(Signature)